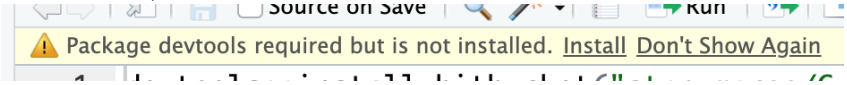
Mat-Bio Exchange: Phase Plane Analysis and Excitable Media

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1. To analyse sets of coupled ODEs, we will be using RStudio, together with the package Grind.R[1]. Grind.R has been developed by Prof. Rob de Boer, Theoretical Biology/Bioinformatics, Utrecht University, the Netherlands. It has been specifically developed for simulating and analysing ODEs. You can either run RStudio on your laptop (RStudio can be downloaded for free from https://posit.co/downloads/), or you can use Rstudio online (go to https://posit.cloud and create a free account, then open an RStudio instant by selecting "New Project>New RStudio Project"; the next time you log in you can simply directly open your project, by clicking on the name you gave to the project after logging in).

2. First, download the provided .R files to your laptop. If you use RStudio on your laptop, you can simply open the files. If you use the website, go to the lower right panel, choose "Files", choose "Upload", and then select the R files on your laptop. You can then open the R files by simply clicking on them after uploading them.

3. The first time that you open a file which uses Grind.R (or if you have updated R or RStudio), a yellow bar might appear at the top of RStudio that looks like this:


In that case, click on "Install" and wait for the "devtools" (or any other required) package to be installed.

4. Run each code step-by-step by pressing the "Run" button (which performs one step of the code). Please read very carefully all the comments in the file before running each step.

1 Logistic growth

First open "my_first_ODE_model.R". This file is just to understand the basics of defining a model and running it in R, showing how the logistic growth model more or less follows the global human population dynamics.

[1] **GR**eat **IN**tegrator **D**ifferential equations for **R**

2 Lotka-Volterra Predator-prey model

Next, open "sheep_wolves.R". This model describes the most basic predator-prey model as developed by Lotka and Volterra, with parameters realistically capturing sheep-wolf interactions.

1. Go through the code and make sure you understand all the steps.
2. Write down the graphical Jacobian for both equilibria, and determine their stability. Can you describe a serious shortcoming of the model?
3. Start a simulation with no wolves. Can you describe a second serious shortcoming of this model?
4. Modify the model to replace the population growth of the sheep by a logistic growth. Look both at the dynamics and at the phase plane.
5. How many equilibria does the model now have?
6. Again, use the graphical Jacobian to determine stability. Can the non-trivial equilibrium (i.e., the equilibrium for which both the sheep and the wolf population is non-zero) ever become unstable? Do you expect in the long run any oscillatory dynamics, as was previously observed? What does this imply?

3 Paradox of enrichment

Rosenzweig (1971) analysed eutrophication of lakes with algae and zooplankton. In Europe and North America, water pollution was at that time an even larger problem than it is now. For instance, large lakes such as lake Erie were nearly dead. It was well known at that time that the pollution was largely due to the nutrients (nitrogen and phosphate), but it was unclear how such nutrient enrichment of a lake would drive ecosystems to extinction. As always in biology, one can argue for several factors contributing to the negative effects of pollution:

• eutrophication leads to growth of the algae which leads to high concentrations of oxygen during the day, but to low concentrations at night due to oxygen consumption;

• in addition to the eutrophication, water pollution involves toxic waste that kills fish, zooplankton and/or algae;

• eutrophication leads to a shift from edible green algae to toxic blue algae; etcetera.

Summarising, water pollution is a complex mixture of factors that are difficult to untangle. The contribution of Rosenzweig was to develop a solid expectation for the effects of eutrophication by a modelling study focussing on the algae and zooplankton alone. He argued that increasing the carrying capacity of the algae by eutrophication is already expected to destabilise the system (Rosenzweig, 1971). The model that he used and we will analyse is the Predator-Prey model with self-limitation, but now with a Holling Type-II functional response:

$$\frac{dA}{dt} = rA\left(1 - \frac{A}{k}\right) - \frac{cAZ}{s + A}$$
$$\frac{dZ}{dt} = -qZ + \frac{pcAZ}{s + A}$$

1. Make sure you understand all the terms, variables and parameters in the model.

2. Open "paradox.R" and run the model for increasing levels of the algae carrying capacity k. What do you observe? Can you intuitively explain this?

3. Now plot the nullclines, again for increasing levels of the algae carrying capacity k. Can you observe qualitative changes in the way the nullclines cross each other?

4. Using the graphical Jacobian, calculate for the qualitatively different situations the stability of all equilibria.

5. Also plot for different starting points the time dynamics within the phase plane. These are called trajectories.

6. Combining the previous answers, can you now understand and explain the observed dynamics?

7. How would you now explain the paradox of enrichment?

8. Rosenzweig was criticised by McAllister et al. (1972), who was growing salmon in ponds. To increase the salmon production he added nutrients to the ponds to feed the algae. His criticism was that the system remained healthy. The water remained clear, i.e., the density of algae was not increasing, while the zooplankton density increased markedly (8-fold). Does this criticism invalidate Rosenzweig's analysis?

9. Consider an oligotrophic lake with very low nutrient levels. The lake has a low stable density of algae. Invasion of zooplankton is never successful because an algal population at carrying capacity is insufficient for the survival of the zooplankton. Use the same model and sketch the nullclines for this situation.

10. [Extra] Large studies comparing various lakes show that the density of the algae is increasing with eutrophication. This increase is much larger however when there is no zooplankton (Persson et al., 2001). Extend the model with a small density-dependent death for the zooplankton, and determine the nullclines.

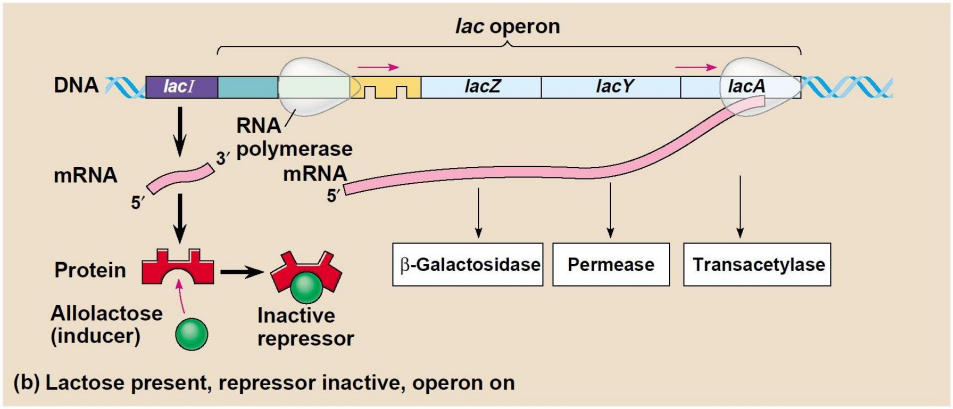
11. [Extra] What do you now expect in this new model for the algal density after eutrophication in the presence and in the absence of zooplankton?

4 The *lac* operon

Bacteria can use several external substrates for cellular growth and switch the corresponding intracellular pathways on and off depending on the available resources. One of the possible substrates is the sugar lactose, and the regulation of the "*lac* operon" was one of the first circuits of gene regulation that was revealed (Jacob and Monod, 1961). Regulation of the *lac* operon involves a positive feedback because the sugar allolactose, A, activates gene expression by deactivating a repressor R.

The disaccharide allolactose is an isomer of lactose that is formed in small amounts from lactose. The intracellular lactose concentration is determined by an enzyme permease that is produced by this gene complex. The gene complex also encodes for the enzyme β-galactosidase that hydrolyses the allolactose into glucose and galactose.

These interactions can be captured by the following model:



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Figure 1: The *lac* operon: regulated synthesis of inducible enzymes.

parameter	value
L	1
c	1
d	1
v	1
c_0	0.05
h	2
n	5
δ	0.2

Table 1: Default parameter values for the model.

$$R = \frac{1}{1 + A^n}\,, \tag{1}$$

$$\frac{dM}{dt} = c_0 + c\,(1 - R) - dM\,, \tag{2}$$

$$\frac{dA}{dt} = ML - \delta A - \frac{vMA}{h + A}\,, \tag{3}$$

where $0 \leq R \leq 1$ is a sigmoidal Hill function representing the fraction of active repressor, M is the mRNA concentration, L is a parameter representing the extracellular lactose concentration, and A is allolactose. The allolactose concentration increases when the gene complex is active, i.e. when permease is present, and extracellular lactose L is transported into the intra-cellular environment. This is formulated as a simple "mass action term" of the external lactose concentration, L, and the permease M. The substrate allolactose is degraded according to a Michaelis-Menten enzyme substrate reaction, with maximum rate v and Michaelis-Menten constant h. This reaction depends on the enzyme β-galactosidase, the concentration of which is again assumed to be proportional to the amount of mRNA M.

1. First look at the model, and make sure you fully understand all variables, terms and parameters. Briefly describe them in your own words.

2. Open "operon.R". The initial set of parameter values is given in Table 1. A good time-scale to run the model is for 40 time units.

3. First analyse the time dynamics of the model. Do this by numerically solving the set of coupled ODEs using Grind.R. Look at the timeplots for both M and A.

What do you observe for different initial values of A (for example, `A=0.0`; `A=0.1`; `A=0.2`; etc.), and what does this imply? Is there any threshold behaviour?

The level of understanding that we can obtain by numerically solving the equations and generating timeplots is limited. It can be further increased by using phase-plane analysis. This basically boils down to plotting along each axis the different variables (variable A along the x-axis and variable M along the y-axis), instead of plotting one variable as a function of time (time along the x-axis and the variables along the y-axis).

4. Start with pencil and paper and draw two axes, with A along the x-axis and M along the y-axis. What is the meaning of a point within this plane? What is the meaning of a "trajectory" in such a plot? Now use Grind.R to generate such a plot. Analyse both using Grind.R and pencil and paper the shape of the nullclines, the structure of the vector field, the stability of the equilibria and the route of the trajectories.

5. Vary the amount of lactose in the environment, parameter L. What happens to the nullclines? What does this imply?

6. Choose a level of L at which there are multiple attractors in the system. Plot the separatrix that separates the different basins of attraction, using the provided code. Make sure you understand what a basin of attraction means.

7. Plot the equilibrium M value and its stability as a function of L, using the provided code.

8. Explain what the generated graph reveals. Can you indicate the lactose levels for which the *lac* operon is 'on', for which it is 'off', and for which it can be either 'on' or 'off' (so-called 'bistability')?

9. After a temporary increase in lactose, will the bacteria always go back to the same state? When is this not the case? In such a case, what will be required to bring it back to its initial metabolic state?

10. The behaviour discussed in the previous question is called hysteresis. Come up with hypotheses as to why such hysteresis could be advantageous (or disadvantageous) for a bacteria.

11. [Extra] In the model above, R, the fraction of active repressor, is represented as a direct function of A rather than an ODE itself. What kind of ODE equations would lead – after for example employing the QSS assumption – to this function? Is a Hill coefficient larger than 1 required for the switching behaviour and what does that imply biologically?

12. [Extra] Explain the experimental results shown in Figure 2, using the above-acquired knowledge. Discuss the cell-to-cell variation that can be observed in both figure panels. Discuss why different levels of cell-to-cell variation can be observed under different experimental conditions.

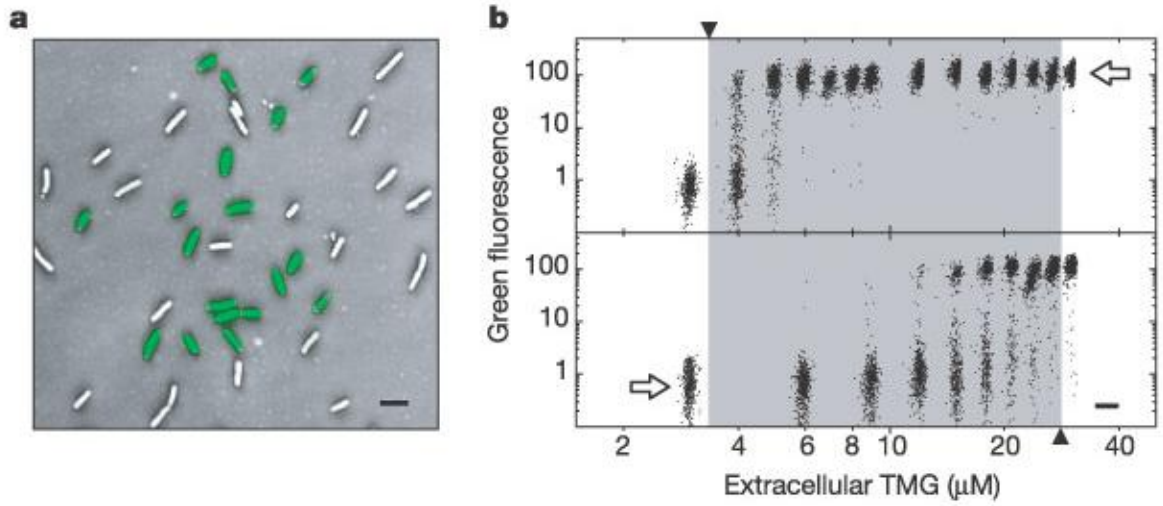


Figure 2: Hysteresis and bistability in single cells. **(a)** Overlaid green fluorescence and inverted phase-contrast images of cells that are initially uninduced for lac expression, then grown for 20 h in 18 μM TMG (which is an analogue of lactose). The cell population shows a bimodal distribution of lac expression levels, with induced cells expressing more than 100x the green fluorescence of uninduced cells. Scale bar, 2 μm. **(b)** Behaviour of a series of cell populations, each initially uninduced (lower panel) or fully induced (upper panel) for lac expression, then grown in media containing various amounts of TMG. Scatter plots show log(green fluorescence) versus log(red fluorescence) for about 1,000 cells in each population. Each scatter plot is centred at a position that indicates the underlying TMG concentration. The scale bar represents variation in red fluorescence by a factor of 10. White arrows indicate the initial states of the cell populations in each panel. The TMG concentration must increase above 30 μM to turn on initially uninduced cells (up arrow), whereas it must decrease below 3 μM to turn off initially induced cells (down arrow). The grey region shows the range of TMG concentrations over which the system is hysteretic. (Figure 2 in Ozbudak et al., 2004)

5 Excitable Media

5.1 Introduction to Excitable Media

Action potentials, very sudden and intense voltage reversals which travel along neurons, belong to the larger overarching class of phenomena known as excitable media. Here you will analyse the dynamics and mechanisms underlying action potential formation and other excitable media phenomena. All 'excitable media' models (a class to which any model of action potentials in neurons belong to as well) share a core set of properties and biological and chemical phenomena that they can capture. Besides action potentials, excitable media models can describe, among others, spruce budworm outbreaks in coniferous forests, the beating of the heart, the life cycle of the cellular slime mould, calcium patterning in oocytes and the Belousov–Zhabotinsky reaction. Although all these processes take place on very different time and space scales, they share very important common features, which you will explore here.

An excitable medium has two main properties, the ability to generate conductible pulses of excitation; and the ability to recover its properties after some period of time, called the refractory period. A classical example is a forest fire. Once enough heat is given to ignite fire in a few trees, the fire (which is a pulse of excitation) is able to spread out through the forest. At a place which is burning, however, the fire cannot sustain itself indefinitely, and once all organic material has been burnt the fire extinguishes by itself at

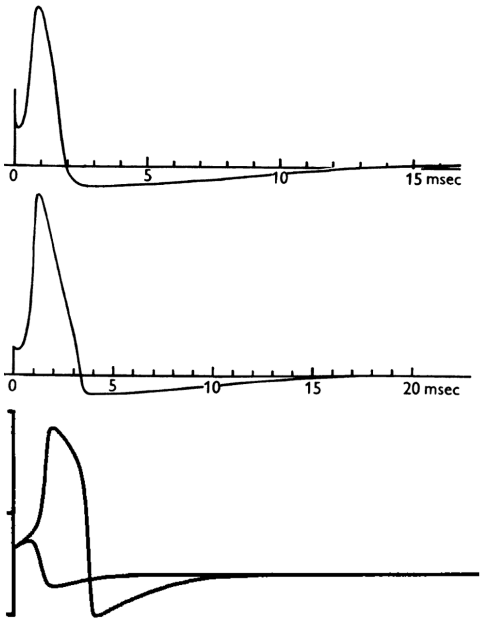


Figure 3: Action potentials. Top row: measured action potential of the giant squid neuron (Hodgkin and Huxley, 1952). Middle row: simulated action potential according to the Hodgkin-Huxley model (Hodgkin and Huxley, 1952). Bottom row: simulated action potential according to the FitzHugh-Nagumo model (FitzHugh, 1961).

that location. Afterwards the forest slowly recovers from the fire (which is the refractory period), until again enough organic material has been formed allowing the possibility of re-initiating fire. It is however difficult to see to which extent such a verbal description of a forest fire helps us to really understand for example action potentials in neurons. More specifically, a forest fire cannot explain why/how sensory neurons are continuously firing while motor neurons only fire when stimulated. The comparison with a forest fire clearly cannot clarify this.

Research on excitable media started with Hodgkin and Huxley who developed in 1952 a model for the generation of action potentials in neurons (Figure 3), a study, for which they received the Nobel prize. Their model was derived from detailed experiments with the giant squid neuron through electrophysiology, and analysed through computational work performed on mechanical calculators. The model consists of a set of coupled Ordinary Differential Equations (ODEs), describing the opening and closing of ion channels and the effect on the membrane potential. All together it was able to closely reproduce a typical action potential.

In 1961, however, Richard FitzHugh (see Figure 4) stated that "The usefulness of an equation to an experimental physiologist (...) depends on his understanding how it works" (FitzHugh, 1961). His criticism was motivated by the belief that such a large set of complex equations is prohibitive to understanding the mechanism driving the observations. He then derived what could be named the core essence of not only neurons but any excitable media in general.

5.2 The FitzHugh-Nagumo model

FitzHugh (1961), and independently Nagumo et al. (1962), developed a simple model that not only captures the main results of the Hodgkin-Huxley model, but of any excitable medium. The model consists of a **set of two coupled ODEs**. The equations are as follows:

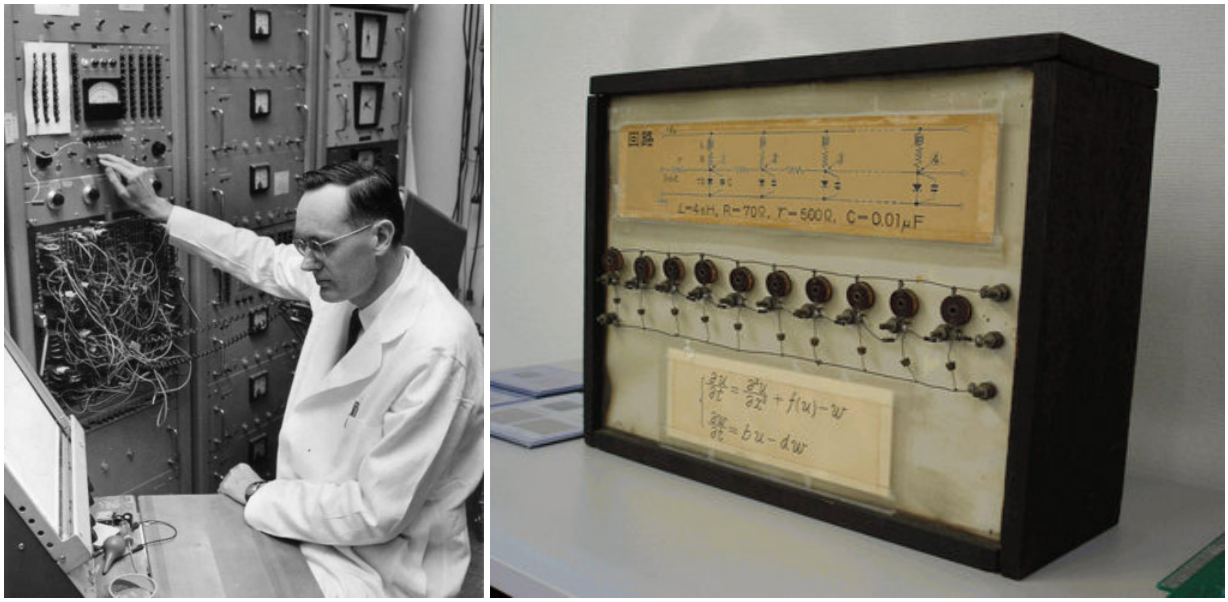


Figure 4: Left: Richard FitzHugh with analogue computer at NIH, ca. 1960. Right: The original Nagumo circuit is held as a trust in the laboratory of Dr. K. Aihara at the University of Tokyo.

$$\frac{dV}{dt} = -V(V-a)(V-1) - W \tag{4a}$$

$$\frac{dW}{dt} = \varepsilon(V - bW) \tag{4b}$$

The variable V is the excitable variable, which can be linked to the membrane potential over the neuron, while the variable W is the recovery variable, linked to ion channel properties. To capture that recovery occurs much slower than excitation, a small parameter ε multiplies the right-hand side in Eq. 4b. With such multiplication, the dynamics of W develop much slower than the dynamics of V.

1. Open the file `fitzhugh.R` in RStudio.

2. The initial values of both V and W are zero. In Grind.R, the ODEs can be numerically solved by using the command run(). Look at the timeplot for both V and W. What do you observe, and what does this imply? Run again, each time for higher initial values of V (for example, by using `V=0.01`; `V=0.02`; etc.). Describe what you observe. Does this resemble action potentials in a neuron? Can you find a threshold?

   The level of understanding that we can obtain by numerically solving the equations and generating timeplots is limited. It can be further increased by using phase-plane analysis.

3. Start with pencil and paper and draw two axes, with V along the x-axis and W along the y-axis. What is the meaning of a point within this plane? What is the meaning of a "trajectory" in such a plot? Now use Grind.R to generate such a plot. Analyse both using Grind.R and pencil and paper the shape of the **nullclines**, the structure of the **vector field**, the stability of the **equilibria** and the **trajectories**. Are there qualitatively different ways to draw the nullclines? Make sure, by reviewing the ODE analysis slides and by discussing with your colleagues, that you understand

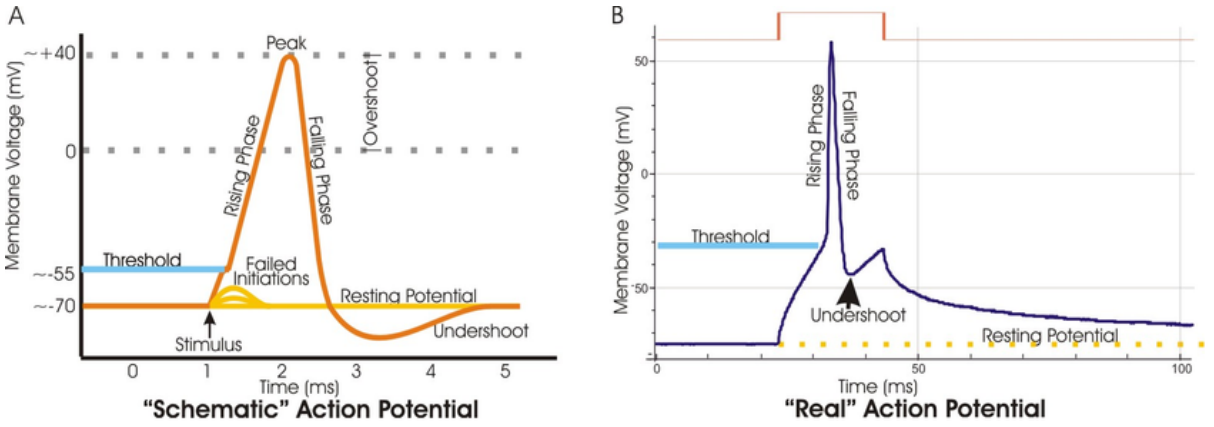


Figure 5: A schematic and real action potential.

what the concepts "nullclines", "vector field", "equilibria" and "trajectories" mean, and how you define them.

4. Start again with different initial levels of V and show the trajectories in the phase plane. What do you observe? How is this related to the timeplots made previously?

5. By looking at the phase-plane, with drawn nullclines and arrows, and taking into account that ε is very small, try to understand the threshold behaviour that was observed in the initial timeplots (question 2). If ε is very small, what would be the value of the threshold? Check this using Grind.R by varying ε.

6. Describe what happens during the generation of an action potential.

7. When an activation/perturbation of equal magnitude is given in the later phase of the action potential, no new action potential is generated. Describe why this is the case, again using the phase-plane. What determines this refractory period?

8. Using Grind.R, if the initial condition is close to the equilibrium, do the dynamics move towards the equilibrium? This will tell you if the equilibrium is "stable" or "unstable". For the observed behaviour, does it make a difference if the parameter a is positive or negative? (Note that in general parameters in biological models are strictly non-negative, but here we make an explicit exemption solely for the parameter a in this model.) Can you understand the resulting dynamics for positive and negative a by looking at the vector field and taking into account that the dynamics of the variable W is much slower?

9. Yesterday you saw that by calculating a "Jacobian" it is possible to mathematically determine if an equilibrium is stable or unstable, and that a powerful "shortcut" to do this, is the "graphical Jacobian". Make sure you again review the slides and remember how this works. Now, determine the stability of the equilibrium for both $a<0$ and $a>0$ by using the **graphical Jacobian**. Note that it might be relevant to assume that ε is sufficiently small.

5.3 The spatial dynamics of the FitzHugh-Nagumo model

Secondly, to run the full spatial simulations, we will be using VisualPDE, a very powerful yet simple to use online PDE solver service.To assume that everything is well-mixed in space is a peculiar assumption for excitable media, as it is essentially about spatial patterns. We therefore should also consider variations in space. To capture the spatial

aspects, we move from ODEs to Partial Differential Equations (PDEs). PDEs differ from ODEs in the sense that the variables also vary in space, and not only in time. In the case of the FitzHugh-Nagumo model there is one essential difference between the ODE version and the PDE version, namely the addition of diffusion to the excitable variable. In two dimensions the model then becomes:

$$\frac{\partial V}{\partial t} = -V(V-a)(V-1) - W + D\left(\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2}\right) \tag{5a}$$

$$\frac{\partial W}{\partial t} = \varepsilon(V - bW) \tag{5b}$$

1. Explain in your own words what $\partial V/\partial t$ and $\partial^2 V/\partial x^2$ mean.

2. Numerically explore the spatial patterns that can be formed by going to the following website: https://visualpde.com/sim/?mini=l8Hcsztg. Here, the variable V is represented by u, and the variable W by v. Initially u is displayed, with the concentrations depicted following the colourbar shown at the bottom, scaled between the minimum and maximum value within the field. To see instead the distribution of v, click on the bottom-left stack symbol and click on v. Note that the colourbar below will then also automatically change to v. Press the "play" symbol on the left to let the simulation evolve. However, if you press "Run" nothing happens. Explain why.

3. Clicking with your mouse in the simulation domain will locally raise V (represented by u) to 1. Trigger a wave by clicking in the upper-left corner. Explain what you see. Next, explore the minimum time required to generate a new wave. Again, explain the origin of this minimum interval. You can reset a simulation by pressing the "return" button on the left.

4. Now, in the simulation introduce a barrier by clicking on the "tool" symbol on the right, then clicking on "Domain" and then on "Custom". What happens when the wave touches the barrier? Once the wave has touched the tip of barrier, what further happens once the barrier is removed by clicking "Custom" again? Explain these results, in terms of the phase-plane and the answer on question 2.

5. Explore what happens when waves collide. This can be done by generating waves wherever you wish, by clicking in the field with your mouse. What are the interesting properties that you notice? Collect and describe them.

6. Now, change the parameters to modify the behaviour of the waves. Parameters can be changed by clicking on the top-left "$f(x)$" button, and then clicking on "Parameters". To remember what the parameters do, you can see the equations at the top of the pop-out panel.

7. Try to understand how you can get faster moving waves, broader waves and larger refractory periods. Try to link these parameters back to what you know from the phase-plane analysis. Also, try to obtain other types of patterns. What do you observe when a is negative?

8. [Extra] Are you able, for negative a, to create within the single domain a spiral wave as well as target waves? This can be easiest done by for example replacing the "Custom Ind. fun" function "abs(y-L_y/2)-10>0 || x>L_x/2" by "(abs(y-L_y/2)-10>0 || x>L_x/4) && (abs(x-L_x/2)-10>0)", and then, to create a spiral on the left-hard-side, by replacing it by "abs(x-L_x/2)-10>0" when the wave hits the barrier.

This should allow for the generation of two independent patterns. What happens when now the barrier is removed?

Suggested Reading

R. FitzHugh. Impulses and physiological states in theoretical models of nerve membrane. *Biophys. J.*, 1:445–466 (1961). The original paper presenting the simplified model.

A. L. Hodgkin and A. F. Huxley. A quantitative description of membrane current and its application to conduction and excitation in nerve. *J. Physiol.*, 117:500–544 (1952). The paper that made them Nobel-price winners.

F. Jacob and J. Monod. Genetic regulatory mechanisms in the synthesis of proteins. *J. Mol. Biol.*, 3:318–356 (Jun 1961). doi:10.1016/S0022-2836(61)80072-7. URL http://dx.doi.org/10.1016/S0022-2836(61)80072-7.

C. D. McAllister, R. J. Lebrasseur, T. R. Parsons and M. L. Rosenzweig. Stability of enriched aquatic ecosystems. *Science*, 175(4021):562–565 (Feb 1972). doi:10.1126/science.175.4021.562. URL http://dx.doi.org/10.1126/science.175.4021.562.

J. S. Nagumo, S. Arimoto and S. Yoshizawa. An active pulse transmission line simulating nerve axon. *Proc. IRE*, 50:2061–2071 (1962). The original paper describing the model.

E. M. Ozbudak, M. Thattai, H. N. Lim, B. I. Shraiman and A. van Oudenaarden. Multistability in the lactose utilization network of *Escherichia coli*. *Nature*, 427(6976):737–740 (Feb 2004). doi:10.1038/nature02298. URL http://dx.doi.org/10.1038/nature02298.

A. Persson, L. A. Hansson, C. Bronmark, P. Lundberg, L. B. Pettersson, L. Greenberg, P. A. Nilsson, P. Nystrom, P. Romare and L. Tranvik. Effects of enrichment on simple aquatic food webs. *Am. Nat.*, 157(6):654–669 (Jun 2001). doi:10.1086/320620. URL http://dx.doi.org/10.1086/320620.

M. L. Rosenzweig. Paradox of enrichment: destabilization of exploitation ecosystems in ecological time. *Science*, 171(3969):385–387 (Jan 1971). doi:10.1126/science.171.3969.385. URL http://dx.doi.org/10.1126/science.171.3969.385.